

Understanding and Avoiding Plagiarism

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Digital Text: Easy Copying, Easy Plagiarism, Easy Detection

- With So much digital contents available in the Internet and highly searchable nature of digital text
 - Easy to detect uncredited copying original documents/materials
 - No of reported incidents has increased steadily since 2005
- Plagiarism means
 - Reuse of someone's prior processes, results, or words without explicitly acknowledging the original author or source.

Plagiarism in any form, at any level, is unacceptable and is considered a serious breach of professional conduct, with potentially severe ethical and legal consequences.

Measuring Plagiarism

- Is there any established measurement that can be used to determine when plagiarism has taken place? In fact plagiarism is plagiarism, but there are degrees of plagiarism:
 - Inappropriate reuse an entire article, or a section of an article, or a page, a paragraph or just a sentence.
 - Even paraphrasing done improperly can be characterized as plagiarism

Plagiarism: Definition

Plagiarism can be defined as an appropriation of the ideas, words, process or results of another person without proper acknowledgment, credit or citation. Plagiarism can appear in a research article or program in following ways.

- a. Claiming another person's work as your own.
- b. Use of another person's work without giving credit.
- c. Majority of someone's contribution as your own, whether credit is given or not.
- d. Restructuring the other works and claiming as your own work.
- e. providing wrong acknowledgment of other works in your work.

Using the Archive: No Exceptions to the Rule

- The fundamental nature of scientific/technical writing is that it is closely based on the archival literature.
- New work typically refers back to any seminal work that has already been published in order to establish a necessary level of authentication and validation.
- If new work depends so much on the close use of the archive, is it reasonable to ask if exceptions should be made for scientific or technical writing where the rules against plagiarism are concerned?
- Some authors believe that technical writings are not literary writing, it should be acceptable to use a “certain amount” of someone else’s text without having to indicate the specific text, especially when a citation appears on the same page or in the reference section.

Using the Archive: No Exceptions to the Rule

- The same kind of thinking might also assert that the use of quotation marks or indented text to identify someone else's text would interrupt the flow of the writing and interface with the reader's comprehension of the work.
- Shouldn't the nature of technical/scientific writing allow exceptions to the formal use of quotation marks and/or indented text

There are no valid exceptions to plagiarism.

Types of Plagiarism

- There are mainly two types of plagiarisms typically found to occur.
 - Textual plagiarism
 - Source code plagiarism.

Types of Plagiarism

The nearest-neighbour based outlier mining technique is able to detect a plagiarized text segment. (*Active voice*)

A plagiarized text segment is detected by the nearest-neighbour based outlier mining technique. (*passive voice*)

(a)

Data: First, Last

Result: Sum

while ($Last \neq 0$) **do**

 Sum=First*Last;

 Last=Last-1;

end

Data: Start, Finish

Result: Total

while ($Finish \neq 0$) **do**

 Total=Start*Finish;

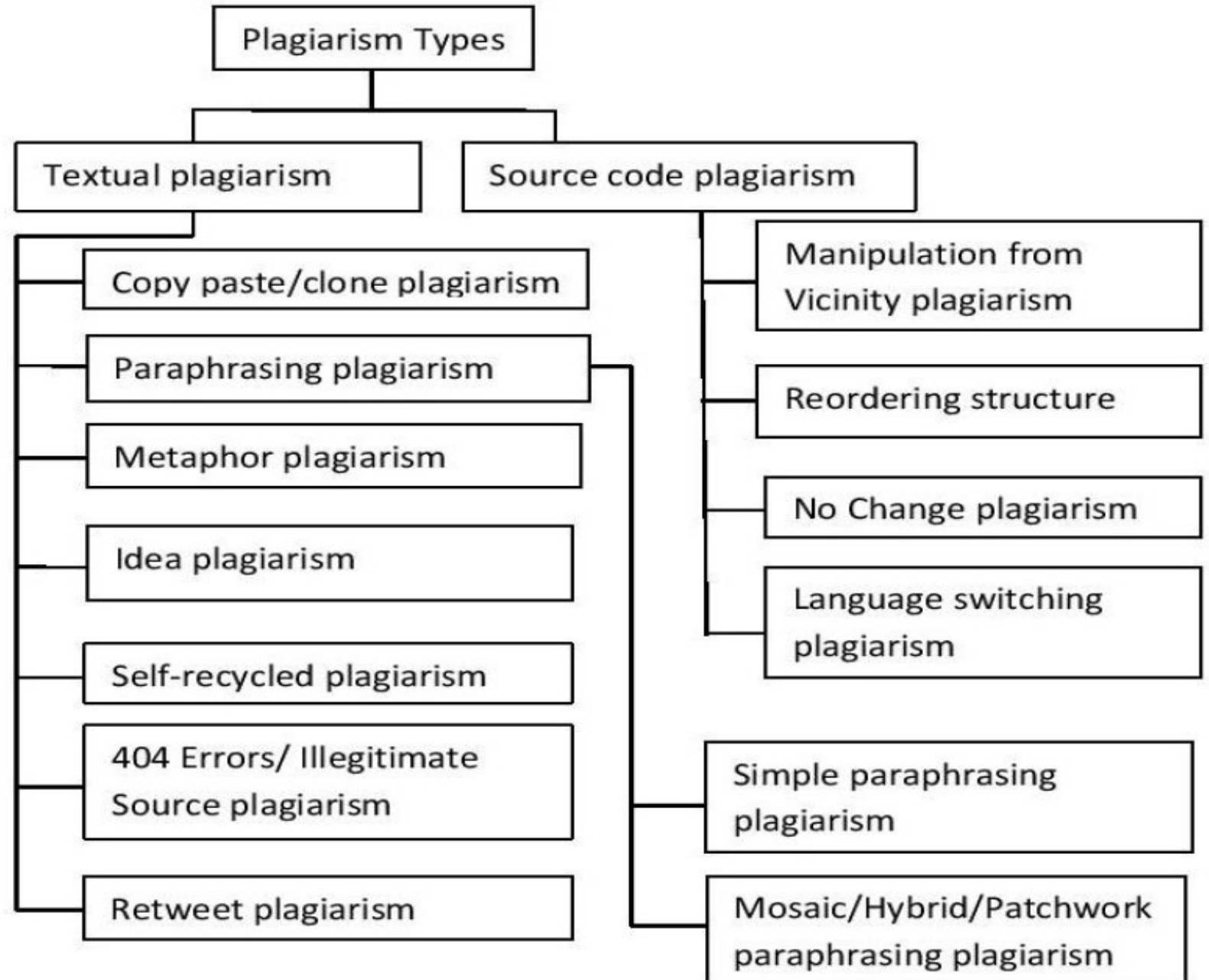
 Finish=Finish-1;

end

(b)

Figure 1: Examples of (a) Textual (b) Source Code Plagiarism

Types of Plagiarism



Types of Plagiarism

Plagiarism can occur between two same or two different natural languages. Based on language homogeneity or heterogeneity of the textual documents being compared, the plagiarism can be of two basic types.

- *Monolingual Plagiarism*: It occurs in homogeneous language settings, e.g., English-English
- *Cross-Lingual Plagiarism*: It occurs in heterogeneous language settings, e.g., English-Chinese.

Textual Plagiarism

Textual plagiarism can be divided into seven sub categories based on its forms and application. We discuss each of these.

1. *Deliberate copy-paste/clone plagiarism*: This type of textual plagiarism refers to copying other works and presenting as if your own work with or without acknowledging the original source.

2. *Paraphrasing plagiarism*: This form of plagiarism can occur in two ways as given below.

- **Simple paraphrasing**: It refers to use of other idea, words or work, and presenting it in different ways by switching words, changing sentence construction and changing in grammar style.
- **Mosaic/Hybrid/patchwork paraphrasing**: This form of textual plagiarism generally occurs when you combine multiple research contributions of some others and present it in a different way by changing structure and pattern of sentence, replacing words with synonyms and by applying a different grammar style without citing the source(s).

Textual Plagiarism

- *Metaphor plagiarism*: Metaphors are used to present others idea in a clear and better manner.
- *Idea plagiarism*: Here, idea or solution is borrowed from other source(s) and claiming as your own in a research paper.
- *Self/recycled plagiarism*: In this form, an author uses his/her own previous published work in a new research paper for publication.
- *404 Error / Illegitimate Source plagiarism*: Here, an author cites some references but the sources are invalid.
- *Retweet plagiarism*: In this form an author cites the reference of proper source but his/her presentation is very similar in the scene of original content wordings, sentence structures and/or grammar usage.

Source Code Plagiarism

This type of plagiarism can be divided into four subtypes.

- *Manipulation from Vicinity plagiarism*: Here, a developer manipulates a program by (i) inserting, (ii) deleting, or (iii) substituting some codes in an existing program, with or without acknowledging the original source and claiming it as his/her own program.
- *Reordering structure plagiarism*: In this type, the developer reorders the statements or functions of a program or changes syntax of a program without referring the original source.
- *No change plagiarism*: Here, the developer adds or removes white spaces or comments or indentation of the program and claims the program as his/her own program.
- *Language switching plagiarism*: In this type, the developer changes the languages, or a program written in one language is rewritten in another language and declare it as his/her own.

How to Avoid Committing Plagiarism

- If author 'A' would like to use text, charts, photographs, or other graphics from author B's original material, then author A must do two important things:
 1. Clearly indicate the reused material by using quotation marks or indented text and provide a full reference to the original material (publication title, author name, article title, etc.)
 2. Obtain written permission from the publisher or, if the reused material has not been published, obtain written permission from the original author.

How to Avoid Committing Plagiarism

- Amount or quantity does NOT play a role in defining plagiarism
- Remember: Plagiarism is plagiarism
- However, the amount of plagiarized materials DOES play an important part in determining the most appropriate penalties when plagiarism is discovered and reported.

How to Avoid Committing Plagiarism

- In general, the appropriate use of credit notices or references are not sufficient to avoid a charge of plagiarism if quotation marks or offset text within the body of an article have not been used to identify the specific material being copied.
- Similarly, improper paraphrasing can leave an author open to a charge of plagiarism if he or she has changed only a few words or phrases or has only rearranged the original sentence order.
- Again, even proper paraphrasing of the original text can lead to a charge of plagiarism if the original source is not properly cited.

A Quick Review

- It is important for all authors to recognize that plagiarism in any form, at any level, is unacceptable and is considered a serious breach of professional and ethical conduct.
- If an author neglects to cite the copied work properly, either by choice or by mistake, the author is committing plagiarism.
- When authors reuse someone else's work and make it appear to be their own, they deny the original author credit for her or his contributions to the research community and to society.
- Plagiarism is a type of *copyright infringement* and as such may also expose an offender to legal liability.

Approach to Combat Plagiarism

A Two-Pronged Approach

- The first is to **detect** plagiarism at the earliest possible opportunity, which is when an article is first submitted.
- The second approach is to **resolve** plagiarism matters as they are discovered after a problem article has already been published.
- Also, as part of plagiarism resolution, one publisher may contact other publishers regarding possible copyright infringement when material is copied and re-published without credit in other journals.

The First Approach: Plagiarism Detection

- Plagiarism detection is done primarily through the use of CrossRef's Similarity Check Service (CrossCheck) by volunteers (Editors/Conference Organizers) when articles are first submitted.
- In addition to this computer-assisted form of plagiarism detection at submission, many instances of plagiarism are also found by reviewers and editors in the early stage of peer review.
- Some users (readers and authors) are also helpful in reporting problems when they encounter plagiarism in an already published work.

The Second Approach: Plagiarism Resolution

- Plagiarism found in any article requires a formal review by the editor of the publication or the organizer of the conference in which the alleged plagiarism was found and, in case involving very serious plagiarism, some penalties may be imposed.
- If, at the end of the formal plagiarism-resolution process, it is determined that the author has in fact inappropriately reused another author's work, some journals add the offending author's name and email address in a Prohibited Author Listed (PAL) which is later made available to all editors for use at the time all articles are first submitted.

Consequences

- If an editor subsequently identifies a banned author while reviewing a newly submitted article, the author's article is straightway rejected.
- Authors will not be permitted to resubmit the rejected article until the banned period has ended.

Effective Plagiarism Detection

- **Similarity Check** is a useful service for detection of plagiarism. A significant advantage of this service over other services is that it includes the indexed full-text content of participating member publishers of this service.
- It not only identifies similarity, but also provides a **similarity score** that helps to establish how much of the work is original and how much is copied from other source(s).
- The necessary feedbacks are provided to editors and conference organizers whenever plagiarism is identified during the review process so that the publishing volunteers can more confidently decide to pursue a formal plagiarism-resolution case or to advise the author about other corrective actions.

Effective Plagiarism Detection

The goal of early plagiarism detection is two-fold:

1. To identify at the earliest possible time an article that contains plagiarized content, thereby ensuring that corrective actions are taken before the article is published.
2. To reduce the number of formal plagiarism cases that occur after publication.

Plagiarism Detection Approaches

There are mainly two types of plagiarism detection approaches available based on whether external resources or references are used or not during plagiarism detection.

- *Intrinsic plagiarism detection* : Where no external references are used.
- *Extrinsic plagiarism detection* : Where external references are used.

Effective Plagiarism Detection

- Articles are uploaded for Similarity Checking upon submission by a number of possible methods.
- Most submissions to some periodicals are submitted to Similarity Check through [ScholarOne Manuscripts \(S1M\)](#) submission platform during peer review.
- Articles submitted to conferences or periodicals that do not use S1M may be uploaded to Similarity Check by another article submission vendor contracted by the conference or periodical, or by the conference organizer using some Similarity Check portal.

The Impact of Plagiarism

Impact on Offending Authors

- Plagiarism is a serious form of publishing misconduct that can have far-ranging and severe consequences for the offending author.
- Anyone who has been found guilty of plagiarizing an original author's work will face one or more of the following.
 - Expulsion from university degree programs
 - Job loss
 - Revocation of advanced degree
 - Damaged reputation
 - Prohibition from future publication

Impact on Original Authors and Readers

- The original authors may take personal offense to the blatant and disrespectful theft of their work, which sometimes represents several years of dedicated effort.
- However, beyond its personally offensive nature, plagiarism also causes logistical problems for many readers, researchers and other users.
- When an undetected plagiarized article is published and enters the archive, readers and especially researchers who discover the article may mistakenly cite the plagiarized article as the original source of valuable information. This in turn, can cause a long-lasting *domino effect* of reference inaccuracy and bibliography error.

Impact on Co-authors

- When plagiarism is found in an article, many societies agree that all authors of the article should be held responsible for the misconduct.
- An author who may be thinking about inappropriate copying of published material should keep in mind that such actions could affect the future career and reputations of co-authors as well.
- It is equally important that the co-author of an article checks for plagiarism before agreeing or submitting.
- If an author's name is on the article, he or she will most likely be held accountable for any plagiarism that is discovered.

Referencing and Citation Guidelines

References & Permissions

- In scholarly writing, it is almost always necessary for authors' research to include other authors' research results, or to reuse portions of their own previously published articles, in order to support the new article being prepared for submission.
- However, it is essential for authors to:
 - Give full credit to any reused, previously published work.
 - Identify clearly in the new article those portions that have been reused from an original authors' article.

Authors must also obtain formal permission from the publishers of the source materials, or obtain written permission from the author if the reused material has not yet been published.

Quotation Marks, Citations & Cover Letters

- Similarly, if authors wish to reuse content from their own previously published articles, they must clearly identify all reused contents by using quotes or indents and provide full citations to the original publication of the article.
- If any previously published or submitted material is used as the basis for a new submission, authors must also state this in a cover letter to the editor and in the text of the article and explain how the new submission differs from the previously published work.

Building References

- Sources of published work that have been used to develop the article should be listed separately at the end of the article in the reference section.
- Generally, the sources must be numbered, and each source in the reference list must be given a separate numbered entry, and should be arranged in the order they have been used in the article.
- If content from a source is used more than once within the article, then separate notations should be made to the same referenced source.

Important Guidelines

- When text from another author's article is reused, the text must be set off by either quotation marks or be clearly indented from the surrounding text and include an in-text-reference number that corresponds to the same number in the reference section of the article where citations to the original source can be found.
- Similarly, when paraphrasing or summarizing material from an original source (including a discussion of ideas, processes, arguments, or conclusions), authors must add in-text numbers that corresponds to the same number in the reference section of the article where credit to the original can be found.
- When reproducing figures, graphics, tables or data, authors must refer to the original source when reusing or revising any information from that source.